

A Data-Driven Approach to Global Air Pollution Analysis and Prediction

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Background

Air pollution is one of the most pressing environmental and public health challenges globally. According to World Health Organization (WHO) data, exposure to air pollution, both indoors and outdoors, is estimated to be the cause of about 6.7 million premature deaths annually worldwide. Contamination of the atmosphere by chemical, physical, or biological agents, such as Nitrogen Dioxide (NO₂), surface Ozone (O₃), Carbon Monoxide (CO), and Particulates (PM_{2.5}), has a direct impact on health, triggering or exacerbating respiratory, cardiovascular, and even cancer diseases. Given their serious impacts, accurate and continuous monitoring of air quality is essential for governments and environmental agencies around the world. To simplify complex air quality data into easily understandable information for the public, the Air Quality Index (AQI) standard is used. AQI represents the overall level of air pollution, which is generally derived from measurements of several key pollutants. The dataset used in this study provides geolocated information on the overall AQI value as well as the AQI value for each component pollutant in various cities. This detailed data provides an opportunity to analyze the quantitative relationship between individual pollutant levels and the aggregate air quality index.

By mid-2025, amidst increasing urbanization and global environmental challenges, resource constraints often prevented extensive and real-time monitoring coverage. Therefore, there is a need for machine learning models that are able to reliably predict the Air Quality Index (AQI) based on more easily measured pollutant data. Predictive models can not only improve the efficiency of monitoring systems, but also serve as an analytical tool to identify which pollutants are the main contributors to air quality degradation in a region. This is crucial for supporting the formulation of effective and data-based environmental policies. Therefore, this research was conducted to address this need by exploring and comparing several regression models to find the most effective approach.

Problem Statement

This analysis aims to build and evaluate machine learning regression models capable of accurately predicting Air Quality Index (AQI) Value, based on data of individual pollutants such as Carbon Monoxide (CO), Ozone, Nitrogen Dioxide (NO₂), and Particulates (PM_{2.5}), along with their respective AQI categories. To achieve this goal, several regression algorithms, including Linear Regression and Random Forest, will be tested to determine which model yields the best predictive performance, where the performance will be measured using evaluation metrics such as

Mean Absolute Error (MAE), Mean Squared Error (MSE), Root Mean Squared Error (RMSE) and R-squared (R^2). Furthermore, this study also aims to analyze the significance of each pollutant (feature importance) to identify the most dominant key contributors in determining the overall AQI value. Thus, the main focus of this problem is to find the most reliable predictive model, which can later be used for air quality monitoring and provide insights on priority pollutants for mitigation efforts.

Solution

To address the problems described, a machine learning-based solution is proposed that is structured in several key stages, from data preparation to model evaluation. This approach is designed to systematically identify the best performing regression model in predicting AQI Value. Here are the steps of the solution that will be implemented:

1. Data Understanding

Data understanding is the initial stage in the data analysis process that aims to gain an in-depth understanding of the characteristics of the data. This stage includes initial data collection, exploration of data structure and attributes, and identification of potential problems such as missing data, duplication, or value irregularities. Through this process, researchers can recognize basic patterns, relationships between variables, and potential information that can be extracted from the data. A good understanding of the data is essential to ensure the subsequent analysis process is valid, relevant, and in line with the research objectives.

2. Data Analysis and Pre-processing

The first step is to prepare the dataset so that it is ready to be used for modeling.

a. Exploratory Data Analysis (EDA)

Analysis will be conducted to understand the distribution of data, identify outliers, and analyze correlation relationships between numerical variables such as 'AQI Value', 'CO AQI Value', 'Ozone AQI Value', and others.

b. Removing irrelevant features

Removing irrelevant features is the process of eliminating variables that do not contribute significantly to the purpose of data analysis or modeling. Irrelevant features can worsen model performance, increase calculation complexity, and lead to overfitting or misleading results. Therefore, the identification and removal of features that do not have a substantial correlation to the target variable or do not have meaningful variability is an important step in the data pre-processing stage. This process aims to simplify the model, improve interpretability, and speed up computation time without compromising the accuracy of the analysis results.

c. Data standardization

Data standardization is the process of transforming the values of variables in a dataset into a uniform scale, usually by transforming them to have a mean of zero and standard deviation of one. The main purpose of standardization is to remove bias caused by scale differences between features, so that each variable has a comparable contribution in the

analysis or modeling process. This process is especially crucial in machine learning algorithms that are sensitive to data scale. With standardization, models can produce results that are more stable, accurate, and can be interpreted more consistently.

d. Encoding Categorical Features

Categorical features such as 'CO AQI Category', 'NO2 AQI Category', and 'PM2.5 AQI Category' will be converted into numerical format using one-hot encoding technique. This process is essential for the regression algorithm to process the data.

3. Data Splitting

The dataset will be split into two parts: 80% for the training set and 20% for the testing set. The variable 'AQI Value' will be set as the target (y), while the other relevant columns will be the predictor features (X). The 'Country' and 'City' columns will not be included in the modeling to maintain the generality of the model.

4. Regression Model Development

Several regression models will be developed and trained using the training data to compare their performance. The models to be tested include: Linear Regression, Random Forest Regressor, and Polynomial Regression.

5. Performance Evaluation and Validation

Each trained model will be evaluated for its performance on test data. The quantitative evaluation metrics that will be used are: Mean Absolute Error (MAE), Mean Squared Error (MSE), Root Mean Squared Error (RMSE), and R-squared (R^2). In addition, 10-fold cross-validation will be applied to assess the stability of the model performance across different subsets of data and serve as the primary mechanism for detecting potential overfitting.

6. Optimal Model Selection

Based on the evaluation results, a comprehensive comparison will be conducted. The model with the lowest MAE and RMSE values and the highest R^2 Square value will be identified as the optimal solution for this case study.

7. Feature Importance Analysis of the Best Model

Once the optimal model has been identified, the next step is to analyze the importance or contribution of each predictor feature. This analysis aims to provide in-depth insight into which pollutants have the most significant impact on AQI Value. This information is valuable to support strategic decision-making in the formulation of pollution mitigation policies, allowing resources to be focused on the most influential factors.

Through this approach, the resulting solution is not only a predictive model, but also a quantitative justification in choosing the most suitable algorithm for this air pollution case study.

Contribution

The study described in this report makes several significant contributions, both from methodological and practical aspects, in the field of air quality analysis. These contributions are relevant for the development of environmental monitoring systems, especially in a region like Indonesia in mid-2025.

The following are the main contributions of this study:

1. **Development of Optimal Predictive Model** The most notable contribution is the identification of Random Forest Regressor as the most accurate performing model to predict AQI Value based on the dataset. With the lowest Mean Absolute Error (MAE) value compared to other models such as Linear and Ridge Regression, this model provides a tested and reliable tool to estimate the overall air pollution level with only pollutant component data.
2. **Basis for Pollutant Monitoring and Analysis** The model developed offers practical benefits to environmental agencies and policy makers:
 - a. **Monitoring System Efficiency:** This model can be used as a support or backup system. If the main sensor for overall AQI fails, AQI values can still be estimated with high accuracy using data from available individual pollutant sensors.
 - b. **Identification of Key Pollution Contributors:** Although not explicitly explored in this notebook, the Random Forest model enables feature importance analysis. This can help identify which pollutants (e.g., PM2.5 or Ozone) have the most influence on the increase in AQI Value at a location, so that mitigation efforts can be better targeted.
3. **Algorithm Performance Benchmark**

This research presents a comparative analysis of the performance of three different regression algorithms on a global air pollution dataset. The results of this evaluation can serve as a benchmark or reference point for future researchers or data practitioners who want to develop more sophisticated models or test other algorithms on the same dataset.
4. **Foundation for Data-Driven Policymaking**

By providing a model capable of predicting air quality and identifying key pollutants, the results of this study provide quantitative evidence for policy makers. This information can be used to formulate more effective environmental regulations, such as setting stricter emission standards for the most influential pollutant sources, both at the city and country level.

Objective

The objectives of this study are specifically described as follows:

1. **Performing Data Understanding**

Examine the global air pollution dataset to understand its basic structure. This included verifying the data type of each column, identifying and quantifying missing values, and obtaining preliminary descriptive statistics to get an overview of the data.
2. **Conduct Exploratory Data Analysis (EDA)**

Comprehensively analyze the dataset to understand the distribution of numerical and categorical variables. In addition, this stage aims to identify correlations between pollutant variables through visualizations such as correlation matrices and pair plots.
3. **Preparing Data for Modeling**

Perform a data pre-processing stage that includes one-hot encoding to convert categorical variables into numerical format. The dataset will then be divided into training data (80%) and test data (20%) for model validation.

4. **Developing and Comparing Regression Models**
Build, train, and compare the performance of six different machine learning models, namely Linear Regression, Random Forest, and Polynomial Regression, in predicting AQI Value.
5. **Evaluating Model Performance**
Quantitatively evaluate each model using standard metrics, such as Mean Absolute Error (MAE), Mean Squared Error (MSE), Root Mean Squared Error (RMSE), and R2 Square on test data. Cross-validation will also be applied to assess the stability and generalizability of the model.
6. **Identifying the Optimal Model**
Determine and recommend the best performing regression model based on the comparative evaluation results, as the most effective and reliable solution for this air quality prediction case study.

Scope of Work

The Scope of Work in this study will include the following:

1. **Data Source**
The analysis exclusively uses the provided global air pollution dataset.
2. **Target Features and Variables**
 - a. The target variable to be predicted is the AQI Value (overall Air Quality Index).
 - b. The predictor features used include CO AQI Value, Ozone AQI Value, NO2 AQI Value, PM2.5 AQI Value, and encoded categorical variables.
3. **Data Pre-processing**
 - a. Perform one-hot encoding on categorical variables to convert them into numerical format.
 - b. Apply feature scaling to normalize the distribution of numerical data, which is a preparatory step for some regression algorithms.
 - c. Split the dataset into 80% training data and 20% test data to train and evaluate the model.
4. **Modelling**
Implementation and performance comparison of six regression algorithms: Linear Regression, Random Forest, and Polynomial Regression.
5. **Evaluation**
Model performance will be measured and compared using the Mean Absolute Error (MAE), Mean Squared Error (MSE), Root Mean Squared Error (RMSE), R2 Square metrics, as well as cross-validation results.
6. **Feature Importance Analysis**
Once the best performing model is identified, a feature importance analysis will be conducted to determine which pollutant variables have the most significant contribution in predicting AQI values.

Literature Review

1. **Regression**

Regression constitutes a statistical method employed to characterize the relationship between one or more independent variables and a dependent variable, formulated through a mathematical model (Suwarsito, n.d.). The primary objectives of regression analysis are to predict values of the dependent variable based on independent variables and to quantify the degree of influence exerted by the explanatory variables.

As defined by Hatta and Adrenanus (2019), the key variables in regression analysis are:

- a. The independent variable (X), alternatively termed the predictor or explanatory variable, represents the factor that influences or induces changes in another variable.
- b. The dependent variable (Y), also called the response variable, denotes the outcome that is influenced by variations in the independent variable.

The fundamental purpose of regression analysis lies in examining the X-Y relationship to identify underlying data patterns and facilitate predictive modeling for new observations.

2. Linear Regression

Linear regression represents a fundamental statistical technique for modeling the linear relationship between a single independent variable (X) and a dependent variable (Y). This method enables the prediction or estimation of Y-values corresponding to given X-values. As Sumari (2021) elucidates, this relationship can be mathematically expressed through the following linear equation:

$$y = \alpha + \beta x \quad (1)$$

Where:

Y = dependent variable (response variable)

α = constant term (y-intercept)

β = regression coefficient (slope parameter)

X = independent variable (predictor variable)

This equation generates a straight line that characterizes the linear association between X and Y. The magnitude of the slope coefficient β reflects the strength of influence that the independent variable exerts on the dependent variable.

3. Polynomial Regression

Polynomial regression serves as an extension of linear regression, designed to model nonlinear relationships between independent and dependent variables. This approach incorporates higher-order terms of the independent variable, enabling the modeling of curvilinear relationships (Malensang, 2012). As Saridi (2015) demonstrates, the general form of the polynomial regression equation is expressed as:

$$Y = \beta_0 + \beta_1 X + \beta_2 X^2 + \dots + \beta_k X^k \quad (2)$$

Where:

Y = dependent variable

X = independent variable

$\beta_0, \beta_1, \dots, \beta_k$ = regression coefficients corresponding to each polynomial term

Polynomial regression proves particularly valuable when analyzing data that exhibit nonlinear patterns. By incorporating successive powers of the independent variable, this modeling approach

achieves enhanced flexibility in fitting complex data structures, consequently improving predictive performance compared to simple linear models.

4. Random Forest Regression

Random Forest Regressor is one of the algorithms within the ensemble learning method, based on the bagging (bootstrap aggregating) approach. This algorithm constructs an ensemble of decision trees on various random subsets of the training data, then produces the final prediction by averaging the outputs of all the trees. This approach is designed to improve predictive accuracy and reduce the risk of overfitting, which commonly occurs in single decision tree models.

$$\hat{y} = \frac{1}{T} \sum_{t=1}^T h_t(x) \quad (3)$$

Where:

$\hat{y}$ = the final prediction result of Random Forest for input x

T = total number of trees in Random Forest

$h_t(x)$ = prediction result of the t th tree against input x

5. MSE

Mean Squared Error (MSE) is one of the most commonly used evaluation metrics in predictive modeling, particularly for regression analysis. MSE quantifies the average of the squared differences between actual observed values and model predictions. A lower MSE value indicates better model performance, as it reflects predictions that are closer to the true values.

6. RMSE

Root Mean Squared Error (RMSE) is a widely used evaluation metric in predictive analysis that measures the accuracy of a model's estimations. RMSE quantifies the average magnitude of the squared differences between predicted values and actual observed values. A lower RMSE value indicates better model performance in predicting the target variable, as it reflects smaller prediction errors.

7. Coefficient Determinations

R-squared (R^2), or the Coefficient of Determination, is a statistical measure that evaluates the goodness-of-fit of a regression model. It quantifies the proportion of variance in the dependent variable (Y) that is predictable from the independent variable(s) (X). The R^2 value ranges between 0 and 1, where:

- $R^2 = 1$: Indicates a perfect fit, meaning the model explains all the variability of the response data around its mean.
- $R^2 = 0$: Suggests that the model fails to explain any of the variability in the response variable.
- $0 < R^2 < 1$: Represents the percentage of the response variable variation that is explained by the model.

8. MAE

Mean Absolute Error (MAE) is a widely used evaluation metric in regression analysis for measuring prediction accuracy. This metric calculates the average absolute difference between the model's predicted values and the actual observed values. Essentially, MAE quantifies the average magnitude of prediction errors without considering whether the errors are overestimations or underestimations, as it uses the absolute value of each prediction deviation. This characteristic makes MAE a robust metric for assessing the performance of predictive models.

9. Cross Validation

Cross-validation is a statistical technique used in predictive modeling to reliably evaluate a model's performance and prevent overfitting. This technique aims to assess how well a model will perform on unseen data by splitting the dataset into several subsets (folds) and iteratively testing the model on each subset.

Methodology

This section outlines the technical approach used to analyze the dataset and develop the predictive model, adhering strictly to the competition guidelines.

Flowchart

The project follows a structured, step-by-step process, illustrated by the following flowchart description:

1. Data Collection & Preprocessing:

- a. Understanding the Dataset Structure: The global air pollution dataset.csv is loaded into a pandas DataFrame. Initial analysis involves examining the data types, column names, and checking for missing values to understand the dataset's integrity.
- b. Data Preprocessing Techniques: This stage involves cleaning the data. We will check for and handle any missing values, likely through imputation (e.g., filling with the mean or median) or removal if the number is insignificant. Feature scaling (normalization or standardization) will be applied to the numerical input features to ensure that all variables contribute equally to the model's performance, preventing features with larger scales from dominating the learning process.

2. Model Development:

- a. Designing the Architecture from Scratch: A Random Forest Regressor model will be implemented. This choice is based on its strong performance in similar prediction tasks noted in the literature review and its nature as an ensemble of simpler models (Decision Trees), which aligns with the "from scratch" philosophy. The model will be built using foundational libraries like scikit-learn, but without using any pre-trained weights or transfer learning.
- b. Setting Hyperparameters: Key hyperparameters for the Random Forest model, such as the number of trees in the forest (`n_estimators`), the maximum depth of the trees (`max_depth`), and the number of features to consider when looking for the best split (`max_features`), will be defined.

3. Training & Evaluation:
 - a. Splitting the Dataset: The dataset will be divided into training and validation (or testing) sets, typically in an 80:20 ratio. The model will be trained exclusively on the training data.
 - b. Model Training: The Random Forest algorithm will be trained on the preprocessed training data. This involves building multiple decision trees, where each tree is trained on a random subset of the data and features.
 - c. Evaluating Model Performance: The trained model's performance will be evaluated on the unseen validation set using standard regression metrics: Mean Absolute Error (MAE), Mean Squared Error (MSE), and the R-squared (R²) score. These metrics will quantify the model's accuracy and predictive power.
4. Computational Efficiency Optimization:
 - a. Measuring Inference Time: The time taken for the trained model to make predictions on the validation set will be measured to assess its computational speed.
 - b. Optimizing the Model: While Random Forest is generally efficient, we will consider the trade-off between the number of estimators and performance. A smaller number of trees can lead to faster computations while potentially maintaining acceptable accuracy.
5. Result Analysis & Report Generation:
 - a. Analyzing and Interpreting Results: The EDA findings and model evaluation metrics will be analyzed. Feature importance scores from the Random Forest model will be examined to understand which pollutants are the most influential predictors of the overall AQI.
 - b. Documenting Findings: All findings, methodologies, and results will be documented in this technical report.

Pseudo Code

The following pseudo-code outlines the core logic for developing the Random Forest Regressor model from scratch, without pre-trained models.

```
// 1. Data Loading and Preprocessing
FUNCTION main():
    // Load dataset
    dataset = LOAD_CSV('global air pollution dataset.csv')

    // Define features (X) and target (y)
    features = COLUMNS(dataset, ['CO AQI Value', 'Ozone AQI Value', 'NO2 AQI Value',
    'PM2.5 AQI Value'])
    target = COLUMN(dataset, 'AQI Value')

    // Handle missing values (e.g., with mean imputation)
    features = IMPUTE_MISSING_VALUES(features, strategy='mean')
    target = IMPUTE_MISSING_VALUES(target, strategy='mean')
```

```

// Split data into training and testing sets
X_train, X_test, y_train, y_test = SPLIT_DATA(features, target, test_size=0.2,
random_state=42)

// Scale features
scaler = INITIALIZE_STANDARD_SCALER()
X_train_scaled = FIT_TRANSFORM(scaler, X_train)
X_test_scaled = TRANSFORM(scaler, X_test)

// 2. Model Development and Training
// Initialize model from scratch (no pre-trained architecture)
rf_model = INITIALIZE_RANDOM_FOREST_REGRESSOR(n_estimators=100,
max_depth=10, random_state=42)

// Train the model
TRAIN(rf_model, X_train_scaled, y_train)

// 3. Evaluation
// Make predictions on the test set
predictions = PREDICT(rf_model, X_test_scaled)

// Calculate evaluation metrics
mae = CALCULATE_MAE(y_test, predictions)
mse = CALCULATE_MSE(y_test, predictions)
r2 = CALCULATE_R2_SCORE(y_test, predictions)

// Print results
PRINT("Mean Absolute Error:", mae)
PRINT("Mean Squared Error:", mse)
PRINT("R-squared Score:", r2)

// 4. Feature Importance
importances = GET_FEATURE_IMPORTANCE(rf_model, feature_names=['CO', 'Ozone',
'NO2', 'PM2.5'])
PRINT("Feature Importances:", importances)

END FUNCTION

```

Result and Discussion

The evaluation of three types of regression models Linear Regression, Polynomial Regression, and Random Forest Regressor reveals notable differences in their performance. The Linear Regression model achieved an R^2 value of 0.9889 on the test data and 0.9882 on the training data, indicating that the model is capable of explaining nearly 99% of the variance in the data. However, the MAE and RMSE values on the test set were 0.0684 and 0.1061, respectively, suggesting a relatively higher prediction error compared to the other models. On the training set, the MAE and RMSE

were 0.0684 and 0.1083.

The Polynomial Regression model demonstrated a significant improvement in performance over the linear regression model. The R^2 values increased to 0.9941 on the test set and 0.9956 on the training set. The MAE and RMSE on the test set were 0.0423 and 0.0777, while on the training set they were 0.0405 and 0.0661, respectively. These results indicate that increasing model complexity through polynomial transformation enables the model to better capture nonlinear relationships between predictor variables and the target, thereby enhancing predictive accuracy.

Meanwhile, the Random Forest Regressor yielded the most optimal results among the three models evaluated. It achieved an R^2 of 0.9982 on the test set and 0.9998 on the training set, with exceptionally low MAE and RMSE values 0.0033 and 0.0422 on the test set, and 0.0010 and 0.0137 on the training set. This high performance demonstrates the model's ability to effectively capture complex patterns and variability within the data.

Overall, these findings suggest that employing more complex models such as Random Forest can substantially improve predictive accuracy compared to simpler linear models. While Linear Regression remains a suitable option when model interpretability and stability are priorities, Random Forest offers superior accuracy and the ability to model nonlinear relationships, making it the preferred choice in predictive tasks involving complex data structures.

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